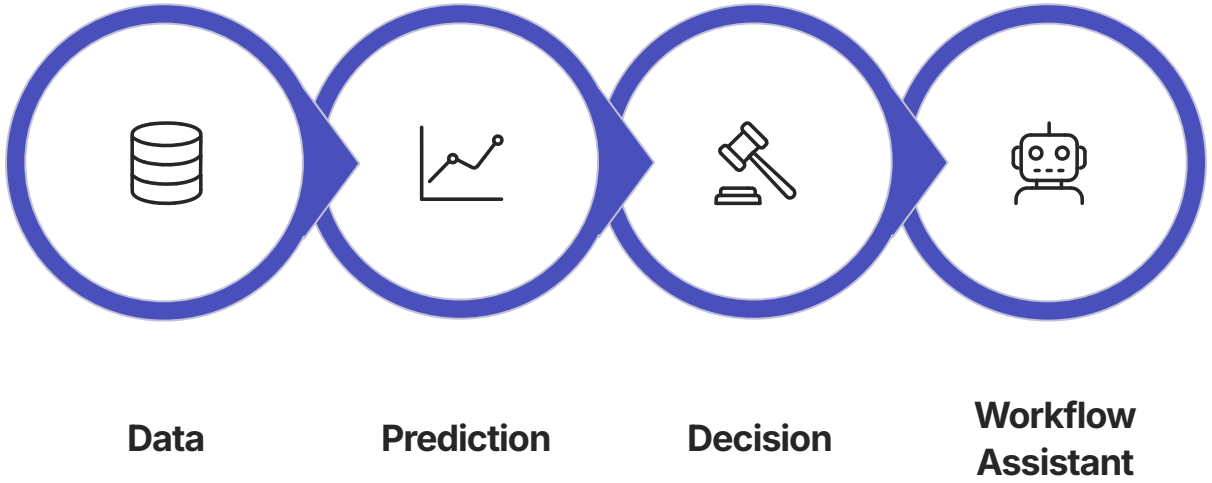
From ML To GenAI In Energy

Today is about understanding where AI fits across **prediction**, **optimization**, and **engineering decision support**.



Session Goal

What You Will Walk Away With

Session 1 should leave participants with two strong takeaways:

Classical ML

Classical ML is useful for **prediction and optimization** in energy workflows

GenAI

GenAI can go well beyond summarization and act as an **analysis and workflow-building copilot**

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- Former Reservoir Engineer and Petroleum Engineering Software Developer
- Built optimization products: WaterWise, SandWise, FracGenie, and Sequestrix
- Experience in production forecasting, decline curve analysis, reservoir modeling, and optimization
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Why This Matters Now

- Technical teams are **drowning in data, reports, spreadsheets, and repeated analyses**.
- Engineering workflows often combine **numerical prediction** and **document-heavy reasoning**.
- AI is valuable when it **saves time, improves consistency, and supports better decisions**.

Production Forecasting

Drilling Report Review

Well-Log Interpretation

Carbon Storage Screening

Three Levels Of Analytics

Understanding where your analysis sits determines which tools and methods apply.

1

Descriptive

What happened? — e.g., production dashboard

2

Predictive

What is likely to happen? — e.g., next-month oil rate forecast

3

Prescriptive

What should we do? — e.g., choose the next surveillance action or operating setting

Where ML Fits

ML is strongest when you have **historical patterns** and want to predict numbers, classes, trends, anomalies, or actions under constraints.

Reservoir & Production

Forecast production, classify underperforming wells

Drilling

Predict risk or performance

Geoscience

Classify facies or lithology

Carbon Storage

Rank sites or estimate capacity

ML Workflow Without The Jargon

Key caution: A good model on bad data is still a bad decision aid.

01

Define the Problem

02

Collect and Clean Data

03

Choose Features and Target

04

Train and Test

05

Evaluate

06

Decide How to Use the Result

Core ML Methods For This Audience

Regression

Predict a number

Classification

Predict a category

Clustering

Group similar wells or intervals

Anomaly Detection

Flag unusual behavior

Time-Series / Lagged Models

Forecast over time

Optimization

Recommend an action under constraints

Role-Specific ML Examples

Production

- Forecast oil rate
- Identify underperformers
- Review lift performance

Drilling

- Predict NPT risk
- Compare offsets
- Flag dysfunction patterns

Geoscience

- Classify facies
- Support log interpretation
- Compare notes across wells

Carbon

- Screen formations
- Rank candidates
- Estimate storage support metrics

Prescriptive Analytics

> Prediction tells you **what may happen**. Prescriptive analytics helps answer **what to do next**.

Optimize Choke or Surveillance Priority

Use prescriptive models to determine the best operating setting in real time.

Rank Wells for Intervention

Prioritize which wells to act on based on expected value and constraints.

Choose Candidate Storage Sites

Select optimal storage sites under technical and economic constraints.

Optimization In Energy

Optimization Is Bigger Than Many Engineers Realize

Optimization is about finding the **best feasible decision** under real technical, operational, and economic constraints.

Products Built

- **WaterWise**: optimize completions water sourcing, storage, routing, and reuse across constrained networks
- **SandWise**: optimize proppant logistics and supply-chain decisions under cost and capacity limits
- **FracGenie**: optimize stage sequencing and completions scheduling
- **Sequestrix**: optimize CO2 transport networks, source-sink matching, and pipeline reuse for sequestration economics

Broader Energy Use Cases

- Production network routing and allocation
- Artificial lift and surveillance prioritization
- Drilling and completions scheduling
- Emissions, carbon transport, and sequestration planning
- Portfolio, capital, and infrastructure decisions

Practical Path To Learning Linear Optimization

1. **Learn the Building Blocks**

   Decision variables, objective, constraints

2. **Start Small**

   Begin with small allocation or routing problems

3. **Translate One Workflow**

   Convert one engineering workflow into an optimization model

4. **Solve With a Tool**

   Use **Excel Solver, PuLP, Pyomo, OR-Tools, or Gurobi**

5. **Validate Outputs**

   Check against engineering common sense and field constraints

Use Cases

- Which wells to prioritize
- How to allocate water, sand, or transport capacity
- Where to tie in pipelines or place infrastructure
- How to schedule operations under time and cost limits

Hands-On Demo

WaterWise-Style Optimization

This is what **prescriptive analytics** looks like in practice.

1

Define the Business Problem

Supply water to frac pads at lowest feasible cost

2

Show Input Tables

Sources, pads, transport links, scenario parameters

3

Define the Model

Decision variables, objective, constraints

4

Solve in Python

Using **PuLP**

Inspect Optimal Allocation

Source-to-pad volumes · Total cost · Freshwater use · Produced-water reuse

Test a Scenario Change

Tighten freshwater cap · Reduce route capacity · Increase produced-water target

Transition To GenAI

So where does GenAI fit?

Use ML When You Need To

- Predict
- Optimize

Use GenAI When You Need To Work With

- Language and documents
- Reasoning steps
- Code generation
- Workflow assembly

ML	GenAI
Predicts the number	Helps the engineer **use** the number

GenAI In Energy Is More Than Summarization

Search Across Reports

Search across reports and technical notes

Extract Structured Facts

Extract structured facts from unstructured text

Compare Wells & Scenarios

Compare wells, reports, or scenarios

Explain Model Outputs

Explain model outputs in plain English

Write Python or SQL

Write Python or SQL for analysis

Assemble Workflow Tools

Assemble repeatable internal workflow tools

FEATURED SCENARIO

Production Forecasting Copilot

18

Wells

Across 3 pads

6mo

Forecast Horizon

Forward-looking production forecast

Identify Likely Underperformers

Flag wells that are trending below expected production

Flag Possible Drivers

Such as **water cut** and **uptime**

Produce Visuals and Engineering Summary

Deliver a clear, actionable output for the engineering team

What ML Does In This Scenario

Uses Historical Well and Operating Data

Draws on past production and operational records to build a learning foundation

Learns Patterns Linked to Future Oil Rate

Identifies statistical relationships that predict forward production

Supports Well-Level Forecasting

Generates individual well forecasts rather than aggregate estimates only

Can Help Rank Wells by Expected Underperformance

Prioritizes surveillance and intervention candidates

What GenAI Does In This Scenario

Frames the Business Problem

Writes or Improves Python Analysis Code

Suggests Plots

Interprets Outputs in Engineering Language

Drafts a Summary Memo

Packages the Workflow

Helps package the workflow as a notebook or internal mini-tool

Practical Limits And Risks

Poor Data Quality

A good model on bad data is still a bad decision aid

Changing Operating Conditions

Models trained on historical data may not reflect new realities

Workovers and Interventions

Operational changes can invalidate learned patterns

Hallucinated Explanations or Code

GenAI outputs must be reviewed — they can be confidently wrong

Overtrust in Recommendations

Without engineering review, AI recommendations can mislead decisions

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Suggested Reading: Oil And Gas ML And Optimization

Production Forecasting with ML

- Production Forecasting in Conventional Oil Reservoirs Using Deep Learning
- A Data-Driven Approach to Forecasting Production with Applications to Multiple Shale Plays

Well Logs and Subsurface ML

- Automated Well-Log Processing and Lithology Classification by Identifying Optimal Features Through Unsupervised and Supervised Machine-Learning Algorithms
- Machine Learning for Well Log Normalization

Drilling and Development Optimization

- Large-Scale Deployment of a Closed-Loop Drilling Optimization System: Implementation and Field Results
- Joint Optimization of Well Locations, Types, Drilling Order, and Controls Given a Set of Potential Drilling Paths

Carbon and Infrastructure Optimization

- Embedding Existing Pipelines in Design of CO_2 Transportation Networks for Optimal Sequestration Economics

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Session 1 Wrap-Up

ML Helps Predict and Optimize

GenAI Helps Interpret, Accelerate, and Operationalize

Together, More Useful Than Either Alone

> **Bridge to Session 2:** Next time we go fully into LLMs, RAG, chatbots, agents, coding copilots, and MCP-style tool connectivity.